



Using Minimal Hardware to Set Up a Personal Cloud

Seagate Central is touted as a home media storage device. The author comes across one, and tinkers with it to convert it to a personal cloud by connecting to a Linux workstation and an old router. His route to success in this venture is described in this article.

It is incredible how our habits are changing in such a short time. Mobile data has become very inexpensive and speeds are pretty good with 4G. Hence, for many, the primary way of connecting to the Internet is using the Wi-Fi network of a portable 4G hotspot.

All too often we do not back up our photos or media from the phone, which incidentally has become the primary camera as well. The photos are shared quickly and lie on our phones till these run out of memory. We are even avoiding that by auto-uploading and saving the photos on the limitless and free cloud.

This was brought home to me when my son left an unused Seagate Central with me and I wanted to make it useful. The motivation was also to rely less on the cloud storage, just in case changes in the environment eliminate free storage and we are expected to pay for the service. There is the ever present nagging fear as to how much can we trust the service providers and what are our rights, as the service is provided free.

Seagate Central is promoted as a personal cloud designed to work easily with a router or a Windows machine. I had an old router which was no longer functioning. So, it seemed obvious that the Linux workstation would be able to do whatever was needed to make the setup work.

Connecting the personal cloud to a Linux workstation

The storage unit connects to the router using an Ethernet connection. Hence, physically connecting the workstation and storage is very simple – just plug an Ethernet cable. The storage will expect a DHCP server to set its address. To easily access it, we would like to give it a fixed IP. We can use the dnsmasq server, which is a small utility that can serve as a DHCP server as well. The following statements need to be added to the `/etc/dnsmasq.conf` file, which is very well documented:

```
# If you want dnsmasq to listen for DHCP and DNS requests
only on
# specified interfaces (and the loop back) give the name of
the
```

```
# interface (eg eth0) here.
interface=enp4s0
# Uncomment this to enable the integrated DHCP server, you need
# to supply the range of addresses available for lease and
optionally
# a lease time.
Dhcp-range=192.168.1.50,192.168.1.150,12h
# Always allocate the host with Ethernet address IP Address
dhcp-host=00:10:75:44:03:50,192.168.1.11,seagate
```

The Ethernet interface on the workstation will need a static IP, e.g., 192.168.1.10.

The firewall for the Ethernet interface should have the http service enabled and the following ports open:

```
137 tcp and udp
139 tcp and udp
445 tcp and udp
4301-4302 udp
```

Browsing `http://192.168.1.11` will provide the options for the configuration and initialisation of the Seagate personal cloud storage.

After that, the storage can be accessed using the 'Other Locations' on the file browser in GNOME.

In Dolphin, we need to give the URL `smb://192.168.1.11`. For convenience, an entry may be made on the left panel of Dolphin.

Accessing the personal cloud by using the Wi-Fi hotspot

Since we do not have a router but a portable mobile hotspot, it is not plug and play! When the mobile hotspot is connected to the workstation, it creates a wired Ethernet interface. The connection allows the workstation to access the Internet. Devices, including phones, can also access the workstation in addition to the Internet. The hotspot routing is limited, and does not have the options to route the traffic meant for the personal cloud to the workstation.